

Subject Code	Subject Name	Theory	Practical	Tutorial	Theory	Practical/Oral	Tutorial	Total
IoTCSBCC701	Machine Learning & Blockchain	03	--	--	03	--	--	03

Subject Code	Subject Name	Examination Scheme							
		Theory Marks				Term Work	Practical	Oral	Total
		Internal Assessment			End Sem. Exam				
		Test1	Test 2	Avg. of 2 Tests					
IoTCSBCC701	Machine Learning & Blockchain	20	20	20	80	--	--	--	100

Course Objectives: Six Course Objectives

Sr. No.	Course Objectives
The course aims:	
1	To learn the basic terminologies used in machine learning and preprocessing of data.
2	To learn Feature Selection and various algorithms
3	To learn concepts of Neural Network and Deep Learning
4	To learn key concepts and basics of Blockchain
5	To learn the Consensus mechanism and Smart contracts
6	To learn application areas of Blockchain

Course Outcomes: Six Course Outcomes (Based on Bloom's Taxonomy)

Sr. No.	Course Outcomes	Cognitive levels of attainment as per Bloom's Taxonomy
On successful completion, of course, learner/student will be able to:		
1	Understand Machine learning, Neural & Deep Learning Concepts.	L2
2	Evaluate mathematical parameters and regression concepts towards building efficient models.	L3
3	Discuss architectural paradigms with respect to Neural networks and deep learning.	L2
4	Understand the fundamentals of emerging blockchain technology.	L2
5	Evaluate the different consensus algorithms, and smart contracts while developing solutions.	L5
6	Delineate the New areas of applications for blockchain and machine learning.	L6

Prerequisite: Introduction to Cryptography, Basic Mathematics & Statistics

DETAILED SYLLABUS:

Sr. No.	Module	Detailed Content	Hours	CO Mapping
0	Prerequisite	1. Mathematics: Vectors, matrices, matrix operations, eigenvalues, eigenvectors, Differentiation, integration, Probability theory 2. Statistics: Descriptive statistics, Inferential statistics, Probability distributions	2	--
I	Introduction to Machine Learning	Introduction:- Introduction:- What Is Learning? When Do We Need Machine Learning? Types of Learning, Relations to Other Fields Basic Terminology & Framework:- Machine Learning Terminology Roadmap for building machine learning -- Preprocessing, Training, and Model selection, Evaluating and Predicting Python for machine learning -- Packages for scientific computing, data science, and machine learning Data Preprocessing:- Dealing with missing data, Handling Categorical data, Partitioning a dataset into separate training and test datasets, Bringing features onto the same scale, Select meaningful features Self-learning Topics: Installation and Configuration of development environment for machine Learning.	7	CO1
II	Feature Selection & Algorithms	Feature Selection:- Feature Selection & Filtering, Principal Component Analysis Algorithms: - Regression:- Ridge, Lasso, ElasticNet, Polynomial, Isotonic, Logistic. Naive Bayes:- Bayes' Theorem, Naive Bayes Classifier, Bernoulli Naive Bayes, Multinomial Naive Bayes, Gaussian Naive Bayes Decision Tree & Ensemble Learning:- Binary decision tree, Random Forest, AdaBoost, Gradient Tree Boost. Clustering: - K-Means Clustering Self-learning Topics: Compare the different algorithms for accuracy, precision, recall, F1-score, etc	9	CO2
III	Introduction to Neural Network and Deep Learning	Introduction to Neural Network:- Basic Architecture of Neural Networks for Perceptron and Multi-Layer Neural Network, Training Neural Network with Backpropagation, Issues in Neural Network Training Introduction to Deep Learning:- Artificial neural Network, Deep architecture, Brief introduction to Tensor Flow Self-learning Topics: Training Deep Neural Networks and	7	CO3

		Issues		
IV	Introduction to Blockchain	Introduction:-- History of Blockchain, What is Blockchain?, Centralized v/s Decentralized System, Layers of Blockchain, Advantage of Blockchain Blockchain Foundation:- Cryptography -- Symmetric, Asymmetric, Hash function, Game Theory -- Nash Equilibrium, Prisoner's Dilemma, Byzantine Generals' Problem, Zero-Sum Games, Trees - Merkle Trees Self-learning Topics: Explore the data structure used in blockchain and study Information retrieval.	8	CO4
V	Consensus Mechanism & Smart Contract	Consensus Mechanism:- Introduction to consensus protocols, Types of Consensus algorithm (PoW, PoS, PoET), Key privacy challenges of the blockchain Smart Contracts: Introduction to Smart Contracts, Working of Smart Contracts, Decentralized Applications, Challenges in Decentralized Applications. Self-learning Topics: Explore and Compare various blockchain platforms	6	CO5
VI	Application of Blockchain	Blockchain Applications:- Cryptocurrency, Blockchain in Health Care Self-Learning Topics: Research papers referred in Online Reference No. 6 & 7	3	CO6

Text Books:

1. Shai Shalev-Shwartz; Shai Ben-David, MACHINE LEARNING From Theory to Algorithms, Cambridge University Press, 2014
2. Sebastian Raschka; Yuxi (Hayden) Liu, Vahid Mirjalili, Machine Learning with PyTorch and Scikit-Learn, PackT, 2022
3. Giuseppe Bonaccorso, Machine Learning Algorithm, Packt, 2017
4. Charu C Aggarwal, Neural Network & Deep Learning A Textbook, Springer, 2018
5. Bikramaditya Singhal, Gautam Dhameja, Priyansu Sekhar Panda, Beginning Blockchain A Beginner's Guide to Building Blockchain Solutions, Apress, 2018
6. Kirankalyan Kulkarni, Learn Bitcoin and Blockchain, Packt, 2018
7. Sandeep Kumar Panda, Vaibhav Mishra, Sujata Priyambada Dash, Ashis Kumar Pani, Recent Advances in Blockchain Technology Real-World Applications, Springer, 2023

References:

1. Vinod Chandra S.S, Anand Hareendran S, Machine Learning A practitioner's Approach, PHI, 2021
2. [Gaur, Nitin, et al. *Blockchain with hyperledger fabric: Build decentralized applications using hyperledger fabric 2*. Packt Publishing Ltd, 2020.](#)
3. [Mastering Bitcoin: Unlocking Digital Cryptocurrencies, by Andreas Antonopoulos](#)

Online References:

1. Live Demo : <https://andersbrownworth.com/blockchain/>
2. Udemy Course- Machine Learning & Deep Learning in Python & R – https://www.udemy.com/course/data_science_a_to_z/
3. Public github repository with code samples:

<https://github.com/HyperledgerHandsOn/trade-finance-logistics>

4. Hyperledger Fabric - <https://www.hyperledger.org/projects/fabric>
5. NPTEL – Introduction to Machine Learning - <https://nptel.ac.in/courses/106106139>
6. Shah, D., Patel, D., Adesara, J. *et al.* Exploiting the Capabilities of Blockchain and Machine Learning in Education. *Augment Hum Res* **6**, 1 (2021). <https://doi.org/10.1007/s41133-020-00039-7>
7. M. Hassan, J. Chen, C. Zhu and U. Zukaib, "Adoption of Blockchain-based Artificial Intelligence in Healthcare," *2022 5th International Conference on Artificial Intelligence and Big Data (ICAIBD)*, Chengdu, China, 2022, pp. 140-144, doi: 10.1109/ICAIBD55127.2022.9820137.

Assessment:

Internal Assessment (IA) for 20 marks:

- IA will consist of Two Compulsory Internal Assessment Tests. Approximately 40% to 50% of syllabus content must be covered in First IA Test and remaining 40% to 50% of syllabus content must be covered in Second IA Test

Question paper format

- Question Paper will comprise of a total of **six questions each carrying 20 marks** Q.1 will be **compulsory** and should **cover maximum contents of the syllabus**.
- **Remaining questions** will be **mixed in nature** (part (a) and part (b) of each question must be from different modules. For example, if Q.2 has part (a) from Module 3 then part (b) must be from any other Module randomly selected from all the modules)
- A total of **four questions** needs to be answered.